

Activity -1

1. Medical Image Enhancement

1. Which gray-level transformation would you recommend?

Log transformation

2. Why is it suitable?

It enhances dark tissues by increasing low-intensity pixel values while keeping bright bones almost unchanged.

3. What effect will it have on the image?

The dark tissues become brighter and clearer, improving overall image visibility and diagnosis.

2. Night-time surveillance

1. Which transformation would improve the visibility?

Log transformation.

2. Why is it more suitable than negative transformation?

It brightens dark regions and reveals hidden details, whereas negative transformation only inverts pixel values.

3. Predict the appearance of the enhanced image.

The image becomes brighter with clearer objects, making vehicles and people easier to identify.

3. Fingerprint Recognition.

1. Which gray-level transformation would you apply?
Contrast stretching

2. Explain why?

It increases the contrast between fingerprint ridges and valleys, making the patterns clearer.

3. What would if a log transformation were applied instead?

It increases or ~~would~~ brighten the image but would not improve ridge contrast as effectively as contrast stretching.